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Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110.

(An Autonomous Institution, Affiliated to Anna University, Chennai)

B.E. / B.Tech.End Semester Theory Examinations, November / December 2024.

Fifth Semester

Computer Science and Engineering

UCS2501 COMPUTER NETWORKS

(Regulations 2021)

Time: **Three Hours**

Answer All Questions

Maximum: **100 Marks**

K1: Remembering

K2: Understanding

K3: Applying

K4: Analyzing

K5: Evaluating

CO1:	Describe the principles, design, terminology and concepts of the network models (K2)
CO2:	Select the protocols of data link layer and apply error control mechanisms (K3)
CO3:	Choose the core functions of network layer protocols and apply them for data communication (K3)
CO4:	Select the Transport Layer Protocols suitable for transmission of data. (K3)
CO5:	Illustrate the Application Layer Protocols and their basic functionalities (K2).

Part – A (5× 2 = 10Marks)

		KL	CO	PI
1.	Let us assume n devices in the network and solve for the number of cable links required for mesh and star topology.	K2	CO1	1.1.2 1.3.1 1.4.1 13.3.1
2.	Can a collision occur on a switch? Justify your answer.	K2	CO2	1.1.2 1.3.1 1.4.1
3.	Explain how classful addresses are identified.	K2	CO3	2.1.2 2.1.3
4.	What is Silly window syndrome?	K1	CO4	1.1.2 2.1.3 13.3.1
5.	What is role of HTTP in web communication?	K2	CO5	1.3.1 2.3.1 13.3.1

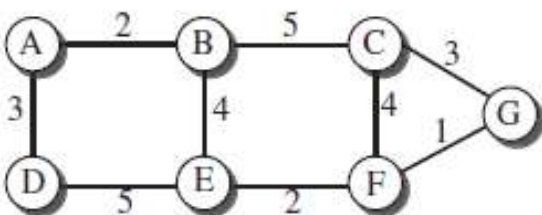
Part – B (5×6 = 30 Marks)

		KL	CO	PI
6.	Explain the TCP/IP protocol suite and its significance in networking.	K2	CO1	1.1.2 1.3.1 1.4.1 13.3.1
7.	A router with IPv4 address 123.45.21.12 and Ethernet address 23:45:BA:00:67:CD has received a packet for a host destination	K3	CO2	1.1.2 1.3.1 1.4.1 2.1.2

	with IP address 124.10.78.10. Identify the entries in the ARP request packet sent by the router. Encapsulate ARP request with the ethernet frame format and show. Assume no subnetting. Also show the entries in the ARP packet sent in response to the router along with ethernet encapsulation.			2.1.3 13.3.1
8.	Describe in detail about the Border Gateway Protocol.	K2	CO3	1.3.1 1.4.1 13.3.1
9.	Explain the Selective Repeat ARQ mechanism for flow control in transport layer with a suitable diagram for corrupted frame and lost ACK.	K2	CO4	1.1.2 2.1.3 13.3.1
10.	Compare and contrast POP3 and IMAP in terms of how they manage email storage and accessibility for users.	K3	CO5	1.3.1 2.3.1 13.3.1

Part – C (5 × 12 = 60 Marks)

		KL	CO	PI
11.	Compute the total delay (latency) for a packet of size 10 million bits being sent over a network with 5 routers, where each router has a queuing time of 1.5 seconds and a processing time of 2 microseconds. The distance between the sender and receiver is 3000 km, the speed of light in the medium is 2×10^8 m/s, and the link bandwidth is 10 Mbps. Identify which component of the total delay is dominant and which one is negligible.	K3	CO1	1.1.2 1.3.1 1.4.1 2.1.2 2.3.1 13.3.1
(Or)				
12.	Develop a socket program in C to implement the simulation of the following using UDP, multiple clients. i. At client side read MAC address from the user ii. Validate the MAC address (size and range). iii. Transmit the MAC address to the server. iv. At server side, find whether the MAC address is multicast or broadcast or unicast address.	K3	CO1	1.1.2 1.3.1 1.4.1 2.1.2 2.3.1 13.3.1
13.	The sender sends the codeword 11110101101. At the receiver side, determine the values for the redundant and data bits at the receiver side by applying the hamming code algorithm. At the sender side introduce error at 6 th bit position and perform error correction.	K3	CO2	1.1.2 1.3.1 1.4.1 2.1.2 2.1.3 13.3.1

(Or)				
14.	Calculate the throughput of the networks that make use of pure and slotted ALOHA random access protocols, if the system (all stations together) produces • 2000 frames per second • 750 frames per second • 300 frames per second The networks transmit 100-bit frames on a shared channel of 100 kbps.	K3	CO2	1.1.2 1.3.1 1.4.1 2.1.2 2.1.3 13.3.1
15.	i) Maggie decides to start a small company. She asks her ISP, Hathway Networks, to give her enough addresses for 205 hosts. Hathway allocates a block starting from 221.240.157.0. Maggie's company has 3 departments, which has its own subnet with host as follows: A: 103 Hosts B: 49 Hosts C: 28 Hosts D: 25 Hosts Design a possible arrangement of subnets to make each department in a different subnet. For each subnet, determine subnet mask and range of IP addresses.	K3	CO3	1.3.1 1.4.1 2.3.1 13.3.1
(Or)				
16.	Apply the distance vector routing for the following network and construct distance vector for each of the node.  Show how the vectors are updated when node A has sent its vector to node B and followed by node E sent its vector to node B. Give the pseudo code of the algorithm and list the limitations of distance vector routing.	K3	CO3	1.3.1 1.4.1 2.3.1 13.3.1
17.	Describe the taxonomy of three-way handshaking protocol used in connection establishment and connection termination of TCP with suitable state transition diagrams.	K2	CO4	1.1.2 2.1.3 13.3.1

(Or)				
18.	Explain the congestion control policies adopted in Reno TCP and how is it different from Tahoe TCP.	K2	CO4	1.1.2 2.1.3 13.3.1
19.	Alex is a system administrator for a tech startup that relies on multiple remote servers for development and deployment. Recently, the team faced a security breach due to inadequate remote access controls. To enhance security, Alex decides to implement the SSH (Secure Shell) protocol for all remote connections. i) Identify the steps Alex should take to set up SSH on the company's main server, including how to configure the SSH daemon and manage user access. ii) As part of the setup, Alex decides to use public key authentication instead of password authentication. Explain the process of generating SSH keys. iii) One evening, Alex needs to troubleshoot a critical application running on a remote server. Illustrate how SSH facilitates this process, ensuring that the connection remains secure, and that sensitive data is protected during the session.	K3	CO5	2.1.2 2.1.3 2.3.1 13.3.1
(Or)				
20.	A traveler named Jamie is sitting at a coffee shop, connected to the public Wi-Fi, and trying to visit a website called "www.goldenfork.com" from his laptop. Illustrate the various processes involved in finding out the IP address associated with the domain name and load the website on the screen. Also, explain how the cache entries are stored in the DNS servers?	K3	CO5	2.1.2 2.1.3 2.3.1 13.3.1